

Personalised AI Assisted Masters Programme Selection

Business and Technical Trade off Analysis | NIU DUOER | PE6201 Section B | 4 October 2026

Final outcome, critique and future path

Problem, user and intended difference

DS Compass is a small evidence-led discovery tool for an international graduate comparing selected master's programmes in Singapore and the United Kingdom. Its primary user has a broad interest, such as artificial intelligence or business analytics, but has to translate that interest into university-specific programme names, English-language rules and official application pages. Today, the authoritative option is manual browsing. That remains necessary, but it is fragmented: a user repeatedly reformulates search queries and then compares differently structured web pages. DS Compass changes only the first research step. It turns a stated direction into a same-domain, source-linked shortlist and makes the English-evidence uncertainty visible. It does not rank universities, predict admission, convert grades or decide degree equivalence.

This boundary follows the original proposal's foundation-model plus official-source retrieval idea while correcting a risky assumption in the early concept: historical applicant cases were not used. The supplied cases did not provide reliable programme-level outcome labels, and their permission, dates and privacy basis could not be established. Treating them as admission ground truth would have made the product appear more personalised while weakening its validity. I instead built a 15-record catalogue from official programme pages and expanded the early Singapore prototype to Singapore and the United Kingdom. The expansion supports cross-country comparison, but the catalogue remains intentionally small enough to inspect record by record.

Final design and decisions

The working path is profile entry, direction selection or short-keyword classification, local catalogue retrieval, English-evidence checking, and result cards with official links. If a user selects one of six fixed directions, retrieval is deterministic. If the user chooses Other or unsure, the worker sends only their short keywords to OpenRouter's Gemini 2.5 Flash Lite and requires one fixed label or abstain. The model is therefore used for a narrow language-variation task, not to generate programme facts. A matching card can only come from the local catalogue. This separation was a deliberate performance and safety choice: an attractive generative answer could easily invent requirements or sound more certain than the evidence permits.

The interface accepts a typed institution and an optional 985, 211 or Double First Class background. These inputs are context only; they do not change the shortlist or create an admission probability. That decision trades personalisation for honesty. In a future system, academic context could support a transparent checklist, but it should not become a proxy score without valid evidence. IELTS overall and component scores are compared only when a card contains a recorded rule. TOEFL is recorded but never converted to IELTS. Where no comparable rule is encoded, the output states review rather than supported. This was a response to an early rough edge: presenting every English result as a pass would have been misleading.

Data, metrics and evaluation critique

`data/programmes.json` contains 15 author paraphrases of official university pages, each with a direct source URL, a verification date, country, domain and explicit English-rule fields. The catalogue's main strength is provenance: a reviewer can follow every card back to its university page. Its weakness is coverage and freshness. Fifteen records cannot represent all suitable programmes, and page content can change. The worker prompts a recheck when a source snapshot exceeds 90 days; this is a reminder, not a guarantee that transcription is correct. A second reviewer and a scheduled refresh process would be more credible than this single-author snapshot.

The LLM evaluation is limited to primary-interest classification. A frozen set of 37 synthetic cases was generated and labelled by GPT-4.1 mini before Gemini was evaluated. Gemini achieved 86.5% exact-label accuracy; the majority-class baseline was 18.9%. The system abstained on 8.1% of cases, and none of those abstentions would

otherwise have been an error against the synthetic labels. These figures exceed the target of demonstrating improvement over a trivial baseline, but they do not measure user benefit, programme suitability, English-rule extraction, eligibility or admissions outcomes. The set is small, synthetic and model-authored, so shared wording and labelling assumptions can inflate performance. I report it as an engineering check, not an independent validation result.

The project also exposed an important tuning trade-off. A permissive classifier would return more recommendations but would force irrelevant directions into the catalogue. A strict classifier would protect users from false matches but abstain too often. The current approach uses fixed labels and an abstain route for unclear or out-of-scope requests. The marine-biology demo deliberately shows that behaviour. The local Laplace-smoothed Naive Bayes fallback improves reproducibility and outage resilience, but it is English-token based and cannot provide comparable Chinese-language performance. This remains a rough edge rather than a solved problem.

Delivery, difficulties and future path

The implementation had three practical difficulties. First, published programme pages use inconsistent wording, so fields such as English thresholds had to be represented as supported, review or gap rather than as a single universal rule. Second, exposing a model key in browser code would have been unsafe, so the hosted worker keeps the key server-side and constrains input length, schema and request rate. Third, the first interface encouraged users to type long goals and did not clearly expose the language-test path. The final interface uses a direction selector, separated IELTS components, a TOEFL option, and an explicit methodology panel. Core tests cover validation, bilingual profile acceptance, English-rule behaviour and abstention; they do not replace end-to-end human usability tests.

The immediate next step is a human-labelled set of genuine, consented discovery queries, split into development and held-out test data before threshold tuning. A second reviewer should verify each programme record and English rule. Later work could add structured prerequisite checklists, broader programme coverage and a user study measuring time saved and whether users reach the official source more efficiently. Those additions should preserve the current constraint: DS Compass can organise evidence for a research conversation, but it should not claim to decide where a person will be admitted.

Conclusion

The project demonstrates a bounded use of external intelligence: language classification assists navigation, while local evidence and explicit rules constrain the result. Its meaningful contribution is not a promise of better admissions outcomes. It is a more transparent starting point than unstructured searching for a small, documented catalogue. The performance numbers are useful for debugging the classifier, but the strongest outcome is the visible separation between recommendation, source evidence, English-rule status and human review.

Sources

Each programme and language source is linked from its corresponding ``data/programmes.json`` record and result card. Model and evaluation details are documented in ``data/metrics.json`` and ``docs/EVALUATION_EXPLAINER.md``.